

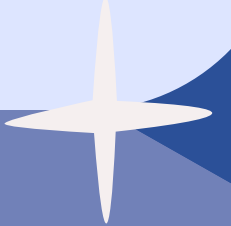


Passive Gravity Compensation **for** Graceful Degradation **of** Hexapod Robots

STRP Check In - 2



PRINCIPLE INVESTIGATOR
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Review – Simulation Results

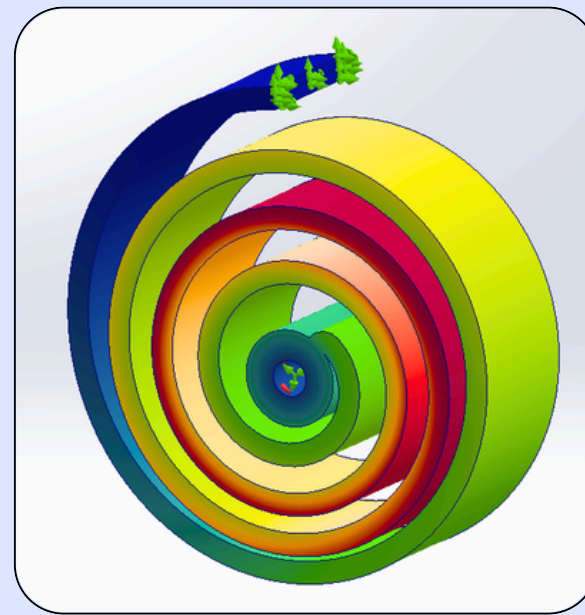


- To have cleaner simulation results fixed the clock time so switched the node to simulated time, so the gait loop and the torque logger both stay locked and had better simulation results.
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Review – Spiral Torsional Spring

Using the Scarcia et al. paper; “Optimal Design of 3D Printed Spiral Torsion Springs” we designed the spring with the set parameters.



We did FEA analysis of the spring to further test our optimised values

Design Parameters



Parameter	Symbol	Value
Desired stiffness	Kd	150 N·mm/rad
Max angular deformation	M	180° (π rad)
Inner shaft radius	RS	12.31 mm
Rod width	b	6.0 mm
Min. number of turns	nm	3
Min. coil gap	p0	2.0 mm
Young's Modulus (PLA)	E	2550 MPa
Yield Stress (PLA)	y	55 MPa
Safety factor		0.75

Optimizer Inputs

Optimizer Outputs

Parameter	Symbol	Value
Spiral rest angle	R	27.4045 rad (4.36 turns)
Rest pitch	pR	6.6771 mm
Pitch at max deformation	pM	5.3799 mm
Rod thickness	t	3.3799 mm
Effective spiral length	Le	328.2046 mm
Rest outer radius	RR	29.1227 mm
Achieved stiffness	K	150.00 N·mm/rad
Stress at max deformation		41.25 MPa (= allowable limit)

Design Parameters



SolidWorks Values

Field	Value
Defined By	Spiral
Pitch	6.6771mm
Reverse direction	Unchecked*
Revolutions	2.518 (= (R - S) / 2 π , live spiral only)



Spring Mount Design



Simulation

AI breathes life into robotics, allowing systems to learn, adapt, and evolve beyond pre-programmed limits.

CoT Analysis



A Vision for Tomorrow

Robotics continues to evolve, inspiring generations to dream, design, and redefine what's possible.

